Project 1: Final White Paper

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Topic

For my first project this term, I propose an interpretable, end-to-end analysis for predicting the presence of heart disease. I will be using a public and well-documented dataset for this task. My goal is to develop a concise, ethical, and actionable modeling workflow that identifies key risk factors and supports practical decision-making, and my hope is to demonstrate a proof of concept for something that can be used for preventative care and early intervention.

Business Problem

Cardiovascular disease is a leading cause of morbidity and mortality worldwide. Health systems and clinicians benefit from simple, trustworthy tools that can stratify patient risk and suggest targeted, non-harmful interventions. The business problem is to estimate heart disease risk at the point of care using routinely collected attributes (for example, chest pain type, resting blood pressure, cholesterol, exercise-induced angina, and ST depression) and to clearly communicate which factors most strongly influence risk. Success is measured both by discrimination on held-out data (accuracy, precision, recall, F1, ROC-AUC) and by the clarity with which the model's reasoning can be explained to clinical stakeholders to inform prevention and follow-up testing plans.

Background/History

This project uses the well-studied UCI Heart Disease dataset (Cleveland subset) curated by Janosi et al. (1989) and hosted by the UCI Machine Learning Repository (Dua & Graff, 2019). The Cleveland subset has become a de facto benchmark in literature because it contains commonly collected clinical attributes and a well-defined target. Prior work often frames the prediction as a binary classification task (presence vs. absence of disease), derived from the original multi-class angiographic disease severity label `num`. The dataset contains 303 records with 13-14 commonly used clinical attributes and a binary target indicating the presence of heart disease. The features include demographics (for example, age, sex) and standard clinical measures (including resting blood pressure, serum cholesterol, maximum heart rate, exercise-induced angina, ST depression, number of major vessels observed by fluoroscopy, and thalassemia). The dataset is available in CSV form and its precise schema will be verified on load. Documented peculiarities will be handled with straightforward imputations and consistent encoding.

Data Explanation (Data Prep / Data Dictionary)

This analysis uses the Cleveland cohort, which contains 304 rows and 17 columns. I define a binary target where a value of 1 indicates the presence of heart disease when the original angiographic label (num) is greater than 0. This yields 165 no-disease cases (54.28%) and 139 disease cases (45.72%).

Data preparation included coercing the fasting blood sugar (fbs) and exercise-induced angina (exang) fields into true booleans, verifying categorical levels for chest pain type (cp), resting ECG (restecg), ST-segment slope (slope), and thalassemia status (thal), filtering to the Cleveland subset, and excluding identifier and label-leakage fields (id, dataset, and num) from the feature set used for modeling.

Key fields include age; sex; cp (typical angina, atypical angina, non-anginal, asymptomatic); trestbps; chol; fbs (boolean); restecg (normal, ST-T abnormality, left ventricular hypertrophy); thalch; exang (boolean); oldpeak; slope (upsloping, flat, downsloping); ca; thal (normal, fixed defect, reversable defect); num (0–4); and the derived binary target.

Methods

I used a stratified train/test split with 25% of the data held out for testing. Preprocessing imputed missing numeric values by the median and standardized them, while categorical and boolean fields were imputed by the most frequent value and then one-hot encoded, with unknown categories ignored at prediction time. I trained two models: a logistic regression as the transparent baseline and a random forest as a non-linear benchmark; both used class-balanced weighting. To avoid leakage, the features excluded id, dataset, and num. Evaluation emphasized accuracy, precision, recall, F1, ROC-AUC, the confusion matrix, and ROC curves, and I also ran five-fold stratified cross-validation for a more stable estimate of generalization.

Analysis

The class balance is modestly skewed toward no disease (about 54% vs. 46%). Age distributions by outcome overlap but suggest that older ages trend toward disease. Chest pain presentation differs by outcome, with asymptomatic cases more common among disease and typical angina skewing toward no disease.

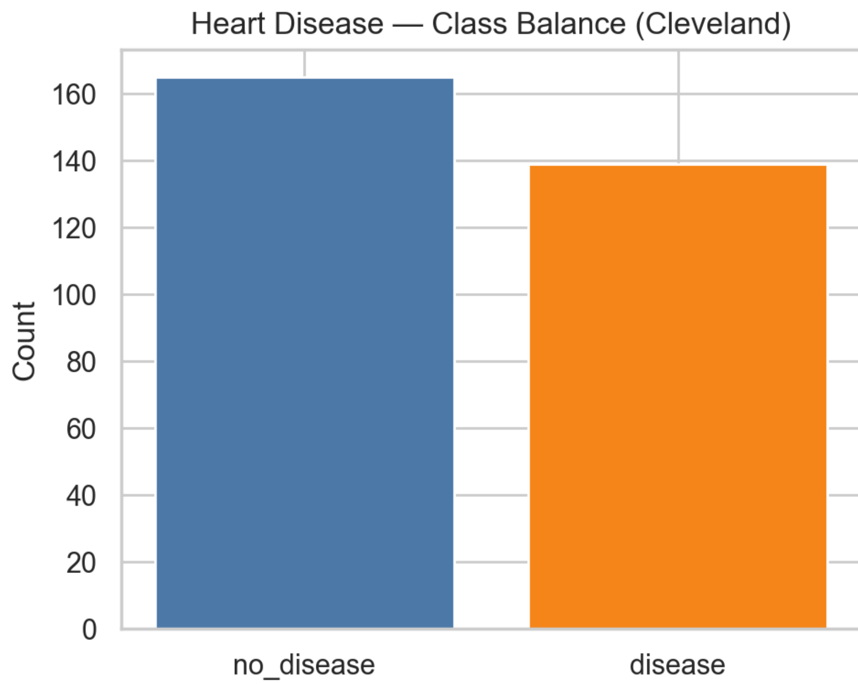

Figure 1: Class Balance

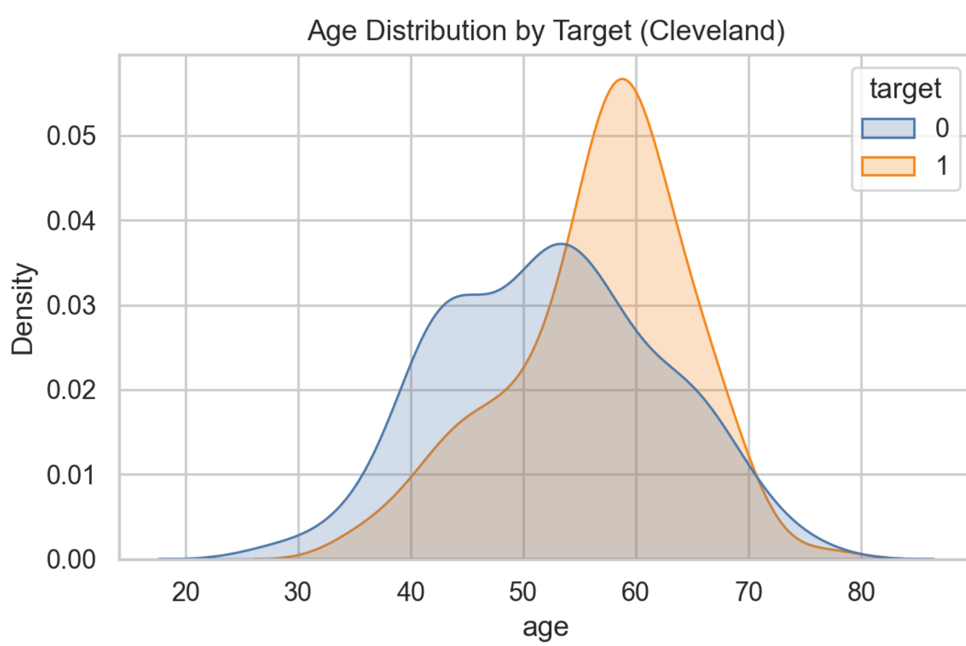

Figure 2: Age by Target

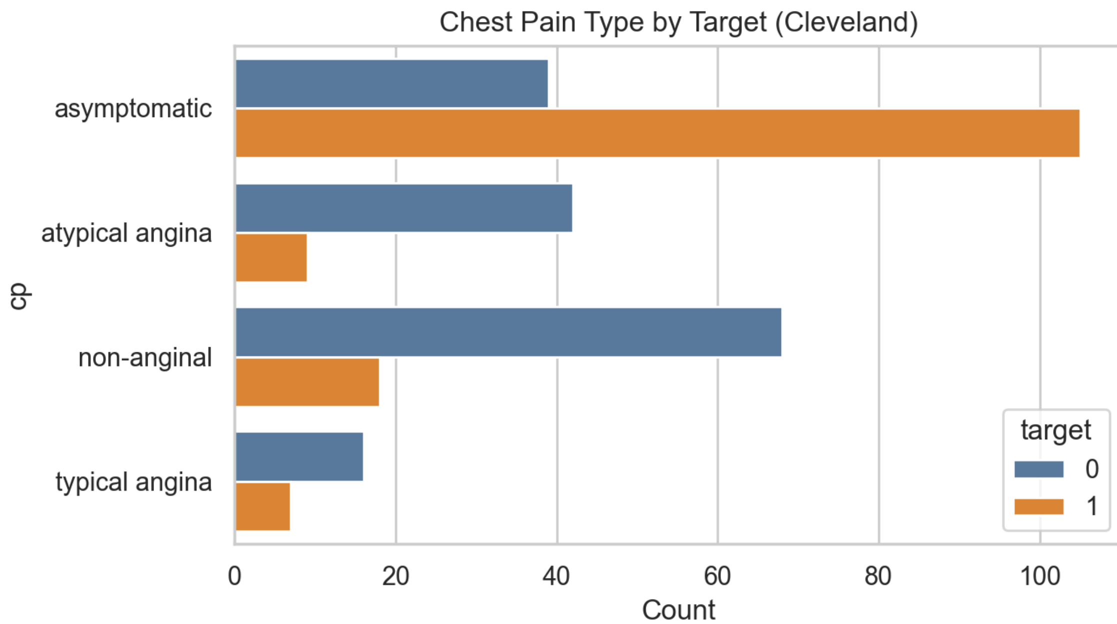

Figure 3: Chest Pain Type by Target

On the held-out test set, the logistic regression achieved accuracy 0.868, precision 0.838, recall 0.886, F1 0.861, and ROC-AUC 0.936, with a confusion matrix of TN=35, FP=6, FN=4, TP=31. The random forest achieved accuracy 0.895, precision 0.886, recall 0.886, F1 0.886, and ROC-AUC 0.952, with a confusion matrix of TN=37, FP=4, FN=4, TP=31.

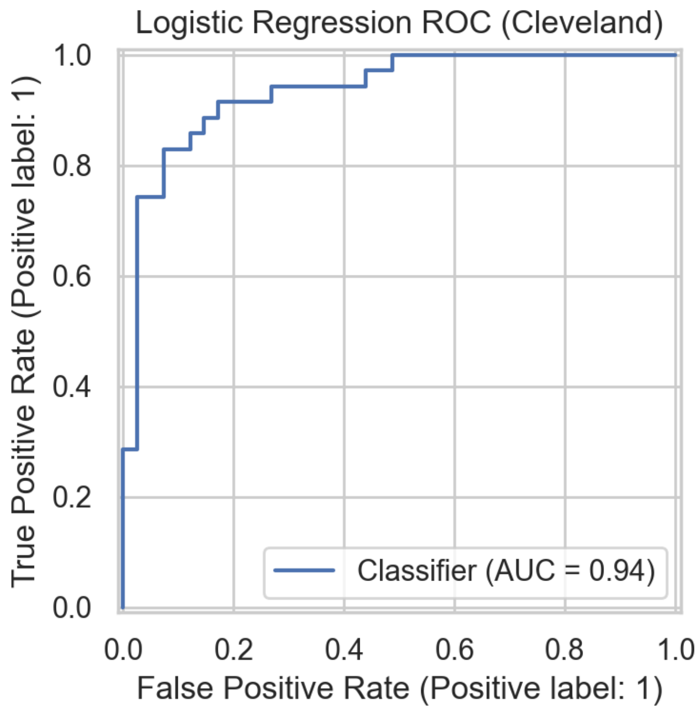

Figure 4: Logistic Regression ROC

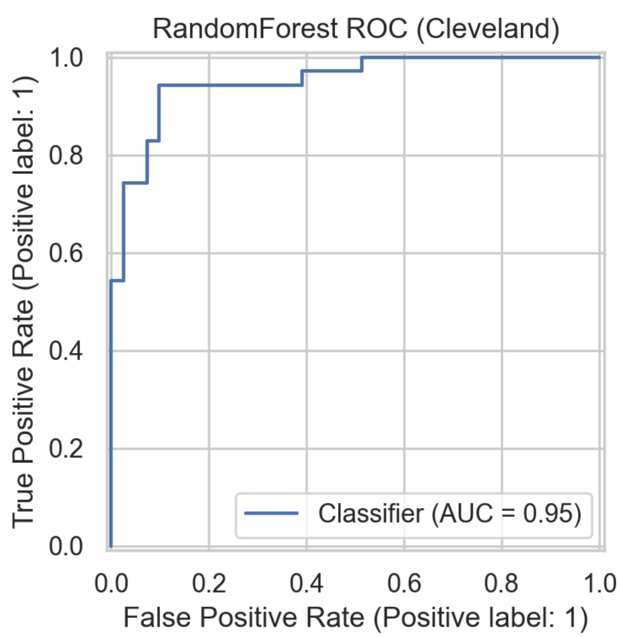

Figure 5: Random Forest ROC

Five-fold stratified cross-validation for the random forest produced mean performance of accuracy 0.829 (±0.027), precision 0.833 (±0.026), recall 0.784 (±0.052), F1 0.807 (±0.033), and ROC-AUC 0.915 (±0.020). I treat these cross-validated values as the primary generalization estimates.

For interpretability, the logistic regression's odds ratios highlight risk-increasing associations for vessel count (ca), asymptomatic chest pain, and reversible thal defects, with protective associations for female sex, typical angina, and higher maximum heart rate. The random forest's permutation importances emphasized ECG and chest-pain markers and cholesterol in this run. These views are complementary and consistent with clinical expectations.

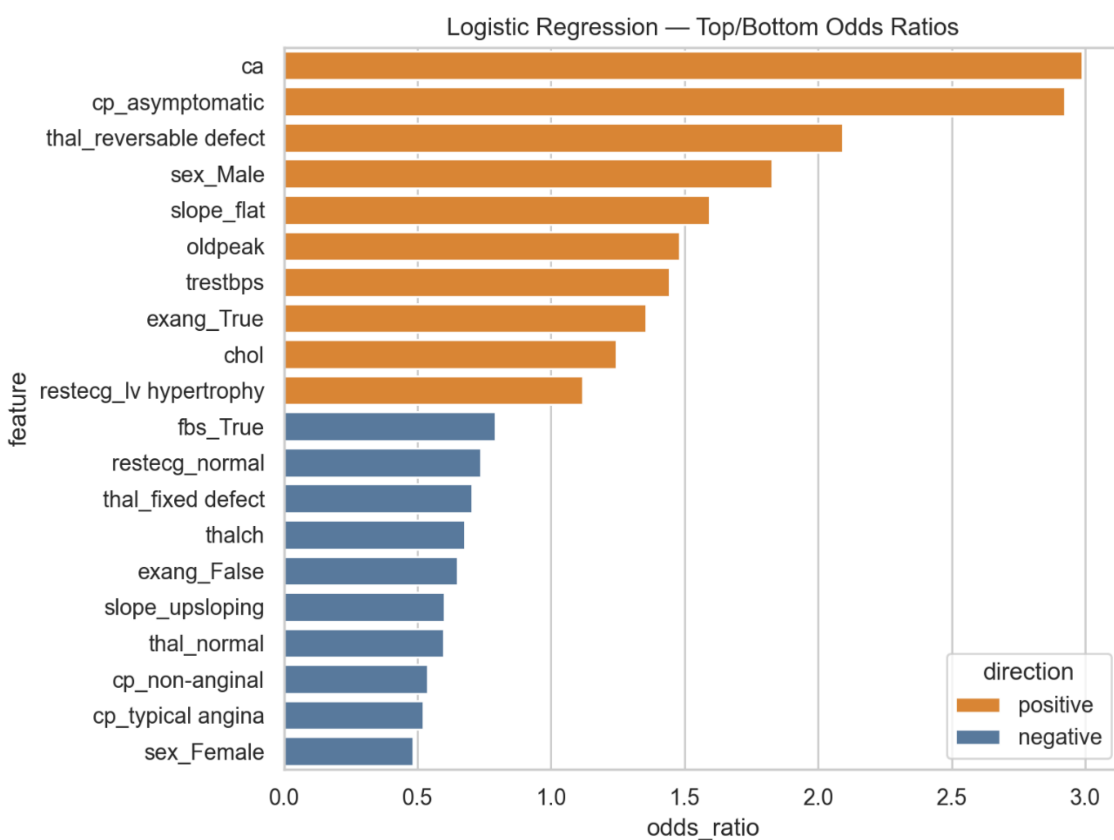

Figure 6: Logistic Top/Bottom Odds Ratios

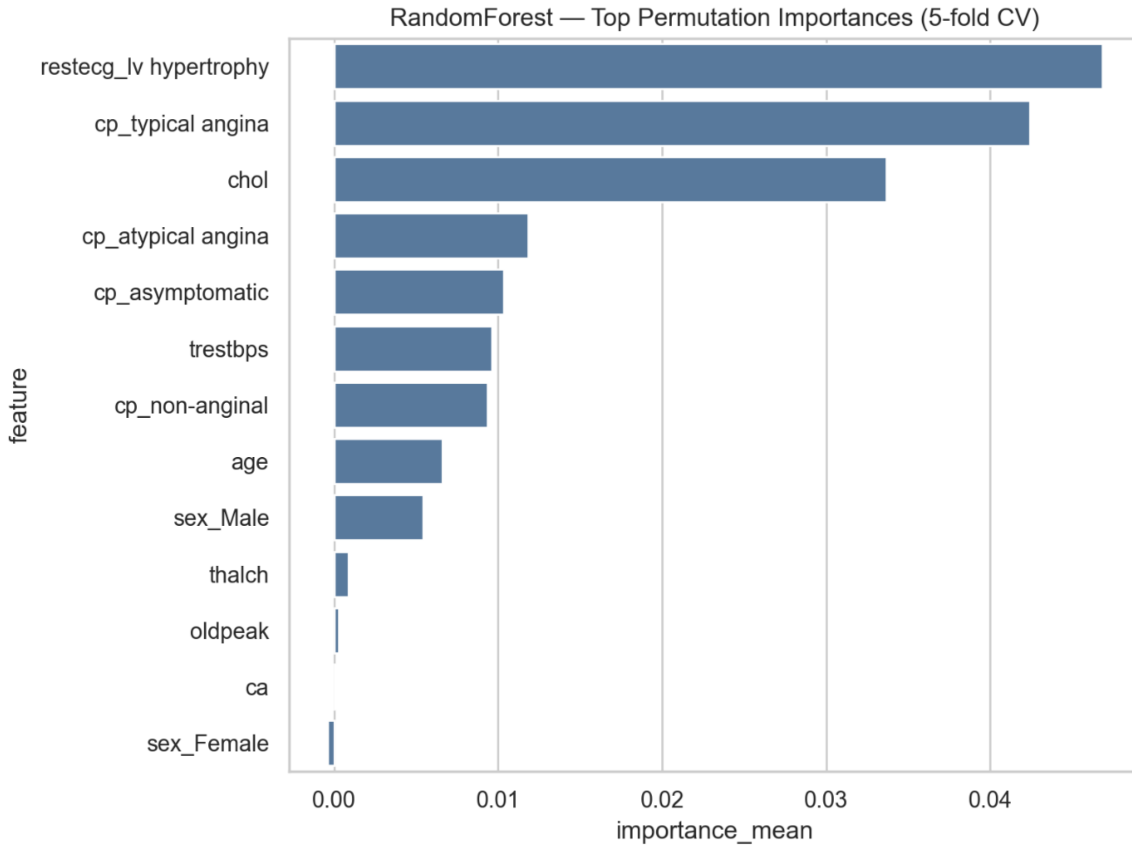

Figure 7: Random Forest Top Permutation Importances

Conclusion

Both models achieved strong discrimination on Cleveland: the RF slightly outperformed logistic on a single holdout split, while cross-validation suggests expected generalization around ROC-AUC ≈ 0.91 for RF. Given the clinical need for transparency, the logistic model is a compelling baseline: it highlights clinically sensible factors (chest pain presentation, vessel count, ST/thal findings, exercise response). RF provides corroboration and potential incremental performance.

Assumptions

I assume the Cleveland subset is a reasonable proxy for a screening population, that missingness is manageable under the simple imputations applied, that one-hot encoding captures the categorical structure sufficiently for baseline modeling, and that class-balanced weighting adequately reflects a practical emphasis on recall without an explicit cost model.

Limitations

The sample is small (n=304), which constrains external validity and raises variance. The data are historical; measurement protocols and prevalence may differ today. Important covariates such as medications and comorbidities are absent, limiting clinical completeness. Finally, I have not yet reported subgroup performance (for example, by sex or age bands), which should be addressed prior to any operational use.

Challenges

Key challenges included handling mixed data types and missingness without leakage, balancing recall with precision to avoid unnecessary follow-up, and reconciling the desire for interpretability with potential gains from non-linear models.

Future Uses / Additional Applications

Potential uses include threshold-tuned risk flags for outpatient triage, decision support for prioritizing non-invasive follow-up testing, and educational dashboards that visualize individualized risk drivers for patients.

Recommendations

I recommend treating the logistic model as the interpretable baseline and reporting calibrated probabilities with clear risk bands. The random forest can serve as a complementary model, provided it is accompanied by permutation-based explanations and partial dependence checks on influential features. Before any pilot, add subgroup performance audits and probability calibration, and establish clinically acceptable thresholds in consultation with stakeholders.

Implementation Plan

The next steps are to package preprocessing and the logistic model into a single inference artifact, add probability calibration and threshold selection informed by decision-curve analysis, and create a simple delivery mechanism suitable for clinicians. I will automate periodic retraining with cross-validation and add monitoring for data and performance drift as well as error analysis. Finally, I will prepare model documentation (model cards) that summarize scope, limitations, and fairness checks prior to any deployment.

Ethical Assessment

This work is intended strictly for educational and exploratory purposes, not as a diagnostic device. I emphasize transparency by prioritizing interpretable models and communicating feature effects with appropriate context. Prior to any use beyond education, I will assess subgroup fairness (for example, by sex and age) and monitor for disparate error rates, and I will avoid deployment if disparities remain unmitigated. Privacy is maintained by using only de-identified public data.

Conclusion

Using the Cleveland cohort, I built an interpretable baseline (logistic regression) and a non-linear benchmark (random forest) that both discriminate well on held-out data (AUC 0.936 and 0.952, respectively) and, for RF, generalize consistently under 5-fold cross-validation (AUC ≈ 0.915). For clinical stakeholders, the logistic model offers a transparent, trustworthy default that surfaces sensible risk drivers (chest-pain presentation, vessel count, ST/thal findings, and exercise response), while the random forest provides corroboration and a modest performance lift. To responsibly operationalize this work, I would calibrate probabilities, select thresholds aligned to clinical costs, audit subgroup performance, and package the pipeline with documentation. With those steps, this workflow forms a practical foundation for risk stratification, prevention planning, and patient education.

This project shows I can use routine, non-invasive information to estimate a person's heart-disease risk and clearly explain the "why." That means clinicians can get a quick, interpretable second opinion about who might benefit from follow-up testing, which factors are most driving risk, and where preventive steps may help. With careful calibration and fairness checks, these models can support better conversations with patients, prioritize limited resources, and help reduce unnecessary procedures while keeping attention on those who need it most.

Appendix (Supporting Documentation)

- The full project code is provided with the submission as a Jupyter Notebook entitled *DSC680_Koyrakh_Milestone_3.ipynb*.
- An audio file presenting this project to stakeholders is attached, entitled *DSC680_Koyrakh_Project_1_Audio.mp3*.
- A supplementary document with anticipated questions and answers from the audience for the audio presentation is attached, entitled *DSC_680_Koyrakh_Project1_QA.pdf*.

References

Dua, D., & Graff, C. (2019). Heart Disease [Dataset]. UCI Machine Learning Repository. https://archive.ics.uci.edu/ml/datasets/heart+Disease

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